

Product Name:Optical Fiber transceiver Module

Brand: YHOPTIC

Model:PCBA-TX1310nm+RX1550nm

Safety Information:

⚠Important reminder: Before flying with fiber, please cut off the fixed cable ties on the fiber storage box, otherwise the fiber will break.

Product Features:

- Surface mounted technology.
- Power supply DC 5.5~26V
- Ferrule Connector(FC)
- Status indicating
- No electromagnetic, frequency and ground current interference;
- Advanced auto-adaptive technology avoiding on-site electric or optical adjustment:

Applications

- It can be applied in the control of unmanned submarines, unmanned aircraft, and unmanned vehicles, for the transmission of images and control signals.

■Module Specification:

Sky Terminal	
Input power	5.5 ~26V
Interface	UART & Video
Video Protocol	PAL/ NTSC
Control Protocol	TTL /SBUS
distance	>60km
Optical wavelength	Tx1270nm /Rx1330nm
Size	62*36mm
Weight	18g
Ground Terminal	
Input power	5.5 ~26V
Interface	UART & Video
Video Protocol	PAL/ NTSC
Control Protocol	TTL /SBUS
distance	>60km
Size	60*30mm

Connecting Application

4. Steps

4.1. Air-Side connecting method

VCC – Connect to the 9V-18V power supply on the flight controller board.

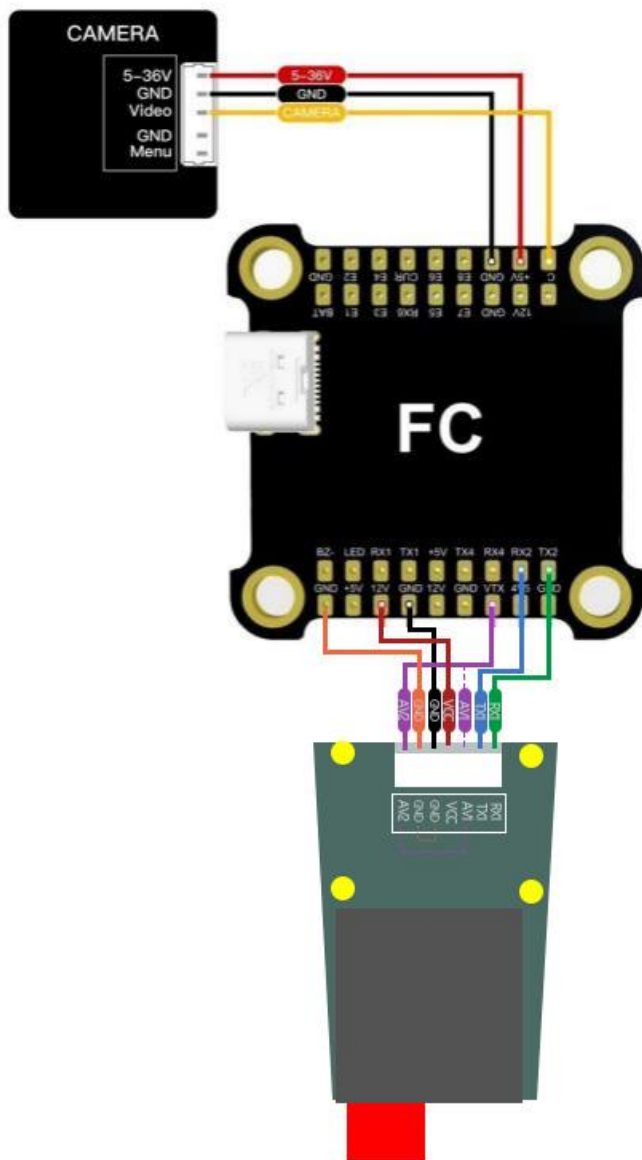
GND – Connect to the GND (ground) pin of the flight controller.

VIDEO – Connect to the VTX (video transmitter) pin on the flight controller.

RX – Connect to the TX pin of the flight controller instead of the receiver

TX – Connect to the RX pin of the flight controller, bypassing the receiver.

FC (Fiber Connector) – Connect to the fiber optic interface at the rear end of the fiber barrel.



Flight Controller (FC)	Sky- Terminal name	function
VTX	AV2	video input
GND	GND	Both interfaces 2 and 3 are GND, and their functions are consistent
GND	GND	
12v	VCC	
VTX	AV1	Video input (AV1/AV2 identical)
RX2	TX1	Flight Control Data
TX2	RX1	Flight Control Data

4.2.Set Up Receiver Port

After connecting the fiber-optic TX module (air-side) to the UART port of the flight controller, note the port number. If connected to UART2, set UART2 as the receiver port in the ground control station (GCS)



4.3.Ground-side Receiver connection method

4.3.1 connect Ground-side Module with ERLS/TBS Receivers

GND → Connect to GND (Ground) pin of receiver

5V → Connect to 5V power supply ($\pm 5\%$ tolerance)

RX → Connect to TX pin (Signal output to receiver)

TX → Connect to RX pin (Signal input from receiver)

4.3.2 Use remote control to match the receiver

ELRS: When using an ELRS receiver with the RX module, to activate binding mode, Rapidly power cycle the ground-side module 3 times, The receiver will then enter binding mode (It will flash green 3 times, then turn off and flash green 3 times again...) - This means that the receiver has switched to binding mode). Next, you need to go to SYS on the control panel and select ExpressLRS

- Bind and Pair, and a connection will be established in a few seconds, the pairing indication will appear on the remote screen and the receiver will start illuminating green statically.

TBS: To pair a TBS receiver, must Activate binding mode; Power on the receiver, The receiver will start flashing green 。 Next , Navigate to the **SYS** menu and select **TBS Agent Lite** 。 Access the Bind tab and Initiate pairing protocol. Connection establishes within seconds , Transmitter displays pairing confirmation indicator, Receiver LED transitions to solid green illumination.

If you have any questions about pairing the receiver with the remote, see Setting up the specifications of the receiver or module on the remote.

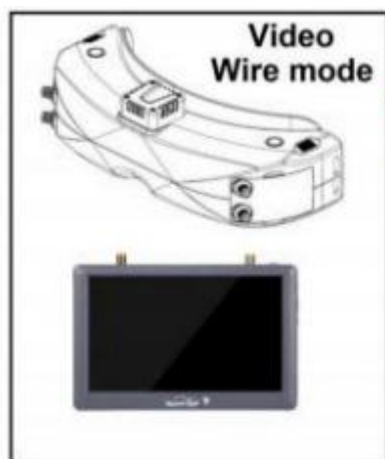
4.3.3 Video Output Connection

A. Wired Video Output

Utilize an AV-to-RCA composite cable for direct video signal transmission to displays or FPV goggles.

B. Wireless Video Output

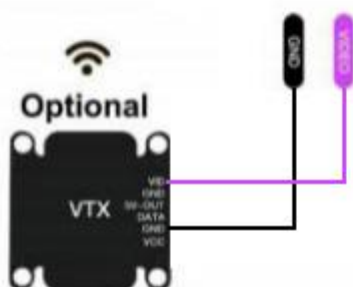
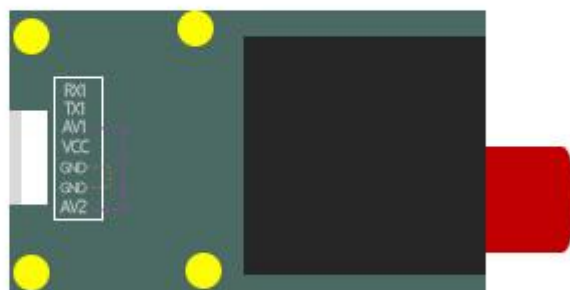
When operating in wireless mode, connect a Video Transmitter (VTX) compatible with all frequency bands.



Video AV out

**RX module
(Ground Terminal)**

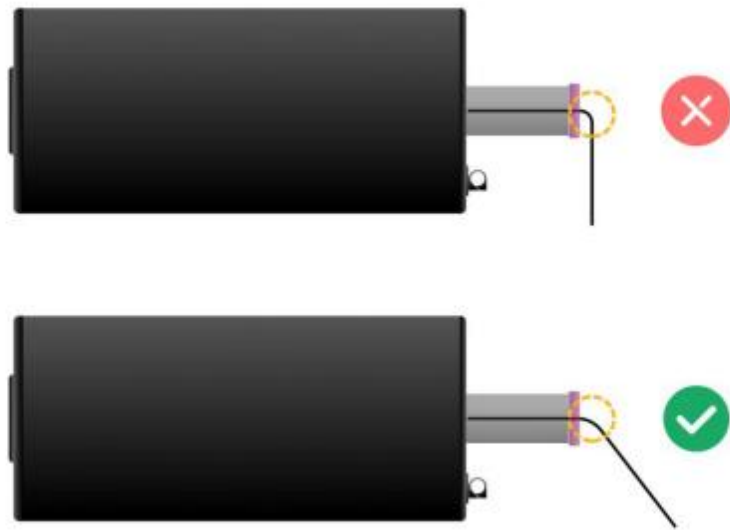
RX1
TX1
AV1
VCC
GND
GND
AV2



Video wire MODE	VTX	ELRS/TBS	Ground-Terminal interface name	function
	Video		AV2	Video output
	GND		GND	Both interfaces 2 and 3 are GND, and their functions are consistent
GND		GND	GND	
			VCC	12V input
AVI			AV1	Video output
		RX	TXI	Serial Flight Control Data
		TX	RXI	Serial Flight Control Data

5. suggestion

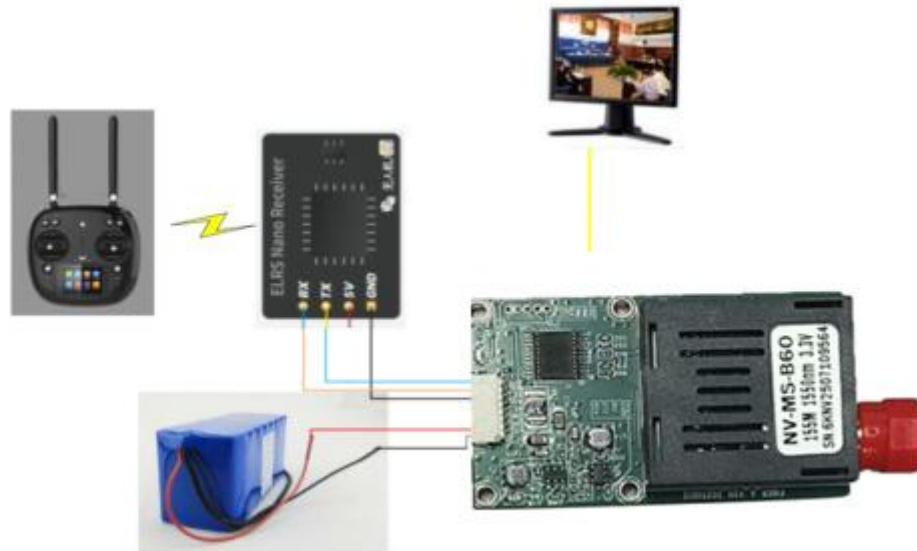
- (1) Keep fiber exit point clear of propellers during flight.
- (2) Avoid abrupt deceleration and rapid descent. Do not descend while turning. Do not turn while descending. These maneuvers may cause the fiber to be severed by propellers.
- (3) Avoid excessive angle when outgoing fiber



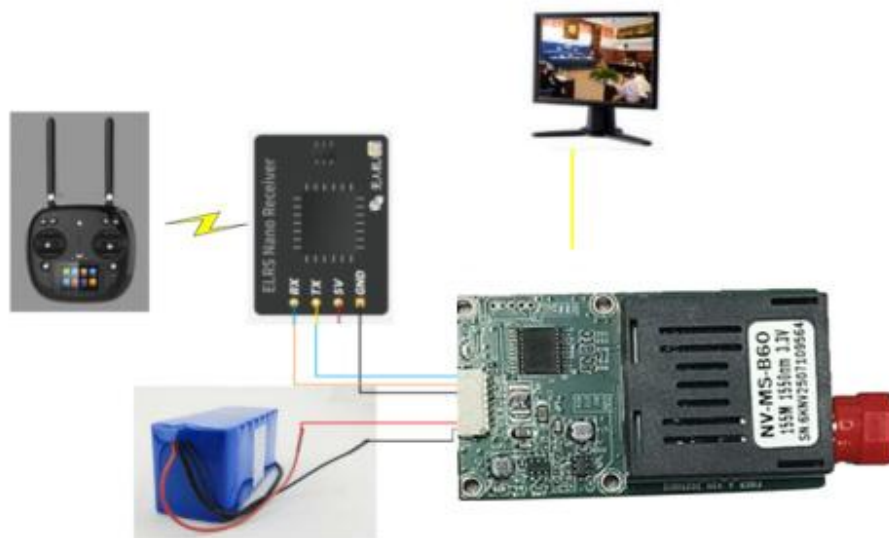
- (4) Limit airspeed to 120 km/h maximum.
- (5) Confirm fiber case latch is unlocked before takeoff.
- (6) Execute gradual acceleration during initial climb, Avoid aggressive acceleration throughout flight, Sudden inertial forces may cause fiber overstrain or connector failure.

■Connection diagram

■Ground-Side Receiver Module (TTL Control)

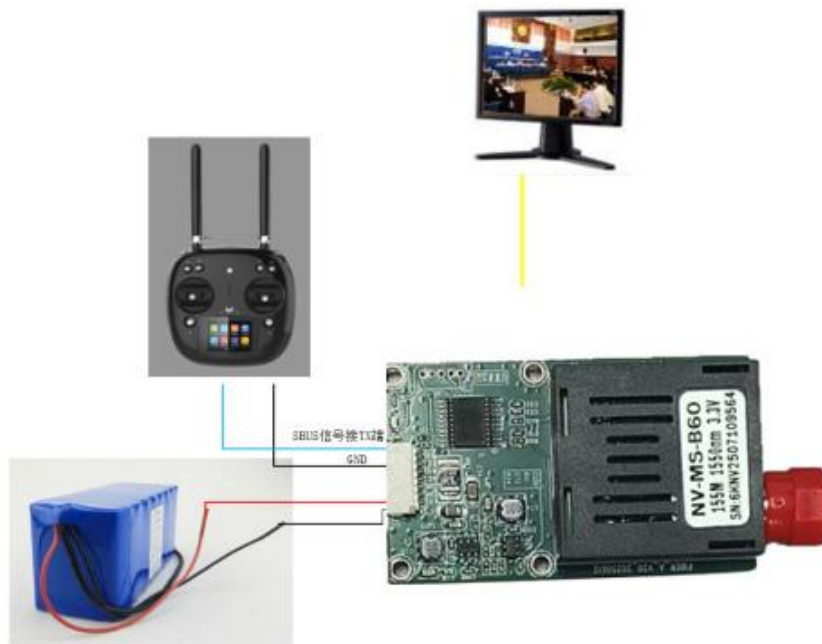


ELRS Receiver to Fiber Optic Ground Station Interface (V1.4)

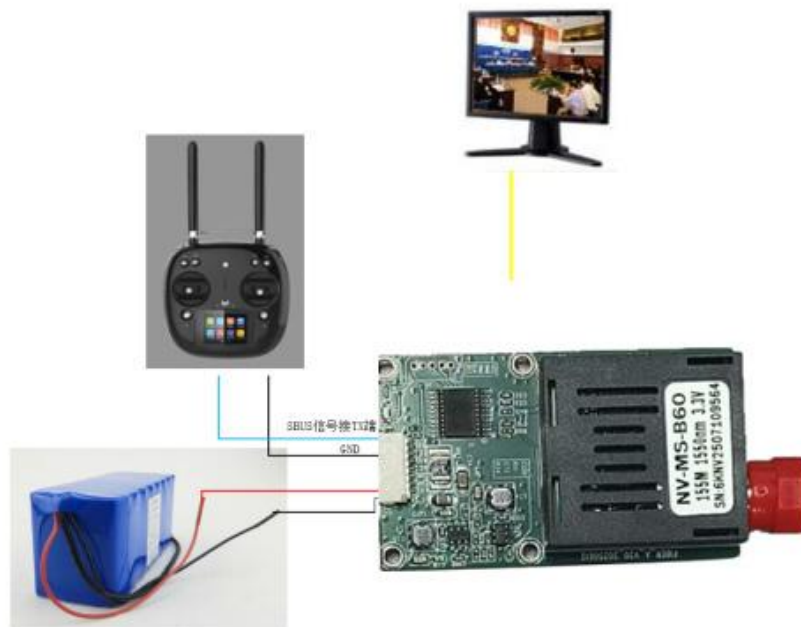


ELRS Receiver to Fiber Optic Ground Station Interface (V2.0)

Ground-Side Receiver Module (SBUS Control)



SBUS Remote Controller to Fiber Optic Ground Station Interface (V1.4)



SBUS Remote Controller to Fiber Optic Ground Station Interface (V2.0)